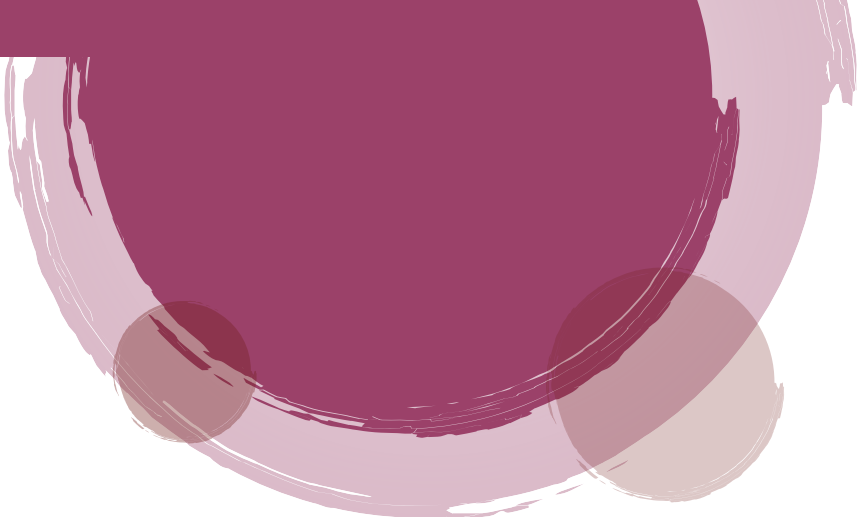




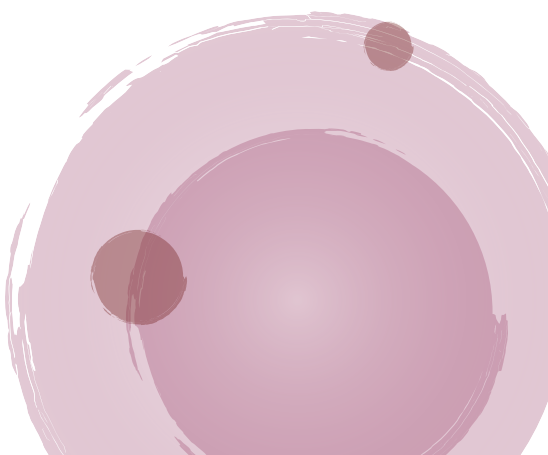
Probability in cybersecurity

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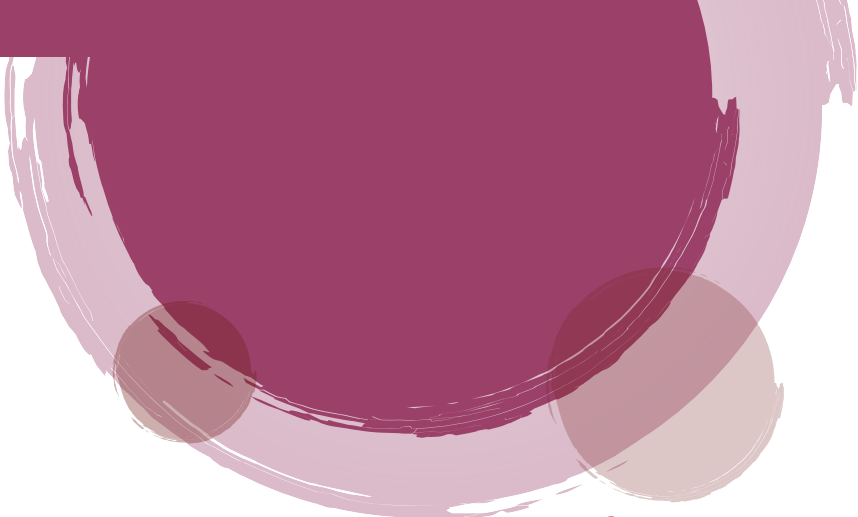
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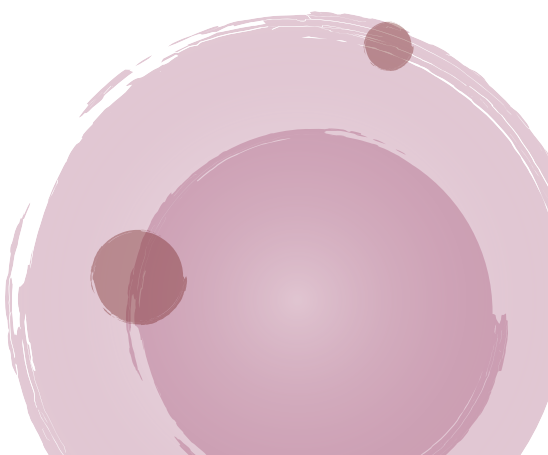
Introduction

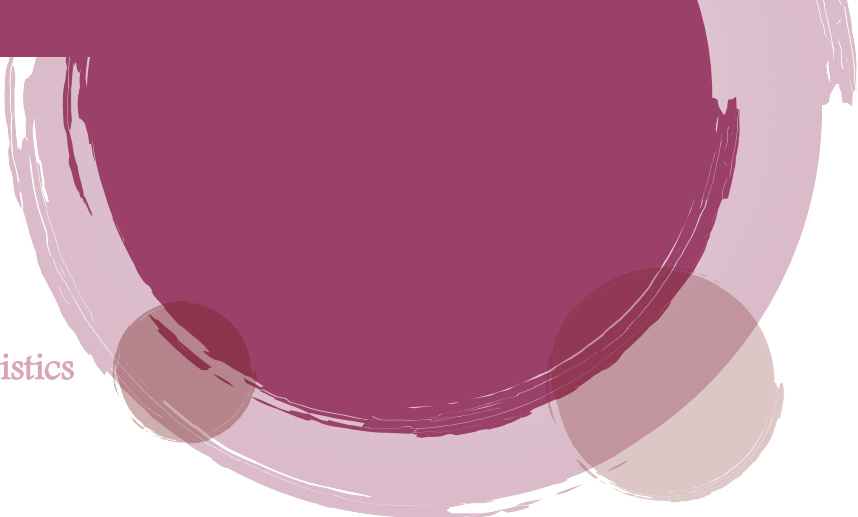


Probability and statistics are among the most important branches of applied mathematics, as they focus on studying and analyzing data, understanding patterns and relationships, and making informed decisions under conditions of uncertainty. This field is distinguished by its ability to transform raw data into meaningful information, helping individuals and organizations predict future outcomes and minimize potential risks.

With the rapid technological development and increasing reliance on digital systems and information networks, probability and statistics have become more crucial than ever, particularly in the field of cybersecurity. These disciplines are used to analyze cyberattacks, understand hackers' behavior, predict potential threats before they occur, and design effective strategies to counter them. They also contribute to improving intrusion detection systems, preventing data breaches, and reducing losses from cyberattacks through quantitative risk assessment and data-driven decision-making.

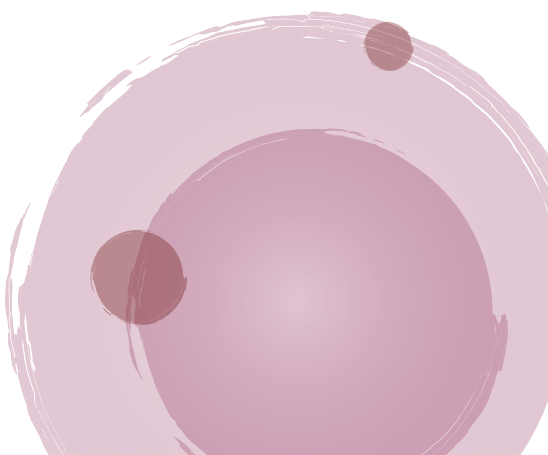
Therefore, it can be said that probability and statistics are no longer merely theoretical mathematical tools but have become a cornerstone in modern technological fields, especially in protecting digital information and securing network infrastructures against increasingly complex and frequent threats.





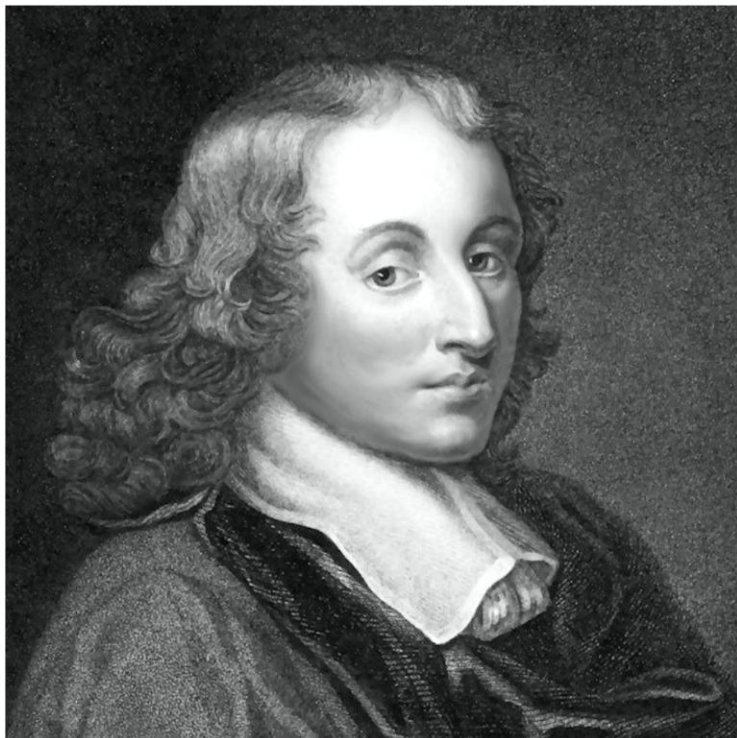
The Origins of Probability and Statistics

developed gradually afterward, with many prominent scientists contributing to the refinement of this discipline and the development of its analytical tools. Among the most notable was Carl Friedrich Gauss, who made significant contributions to the development of the normal distribution and data analysis, providing precise mathematical methods to study natural and social phenomena through the collection and analysis of data. These efforts established statistics as a comprehensive scientific tool capable of understanding patterns and trends and making evidence-based decisions. This made it foundational for many modern sciences, including economics, physics, sociology, and engineering, as well as playing a vital role in fields such as cybersecurity and big data analysis. Statistics, on the other hand, developed gradually afterward, with many prominent scientists contributing to the refinement of this discipline and the development of its analytical tools. Among the most notable was Carl Friedrich Gauss, who made significant contributions to the development of the normal distribution and data analysis, providing precise mathematical methods to study natural and social phenomena through the collection and analysis of data. These efforts established statistics as a comprehensive scientific tool capable of understanding patterns and trends and making evidence-based decisions. This made it foundational for many modern sciences, including economics, physics, sociology, and engineering, as well as playing a vital role in fields such as cybersecurity and big data analysis.





Blaise Pascal



Pierre de Fermat

The Importance of Probability and Statistics in Cybersecurity

Probability and statistics play a crucial role in cybersecurity, leading to:

- Analyze cyberattack behavior: Understand recurring attack patterns and anticipate attackers' moves.
- Estimate the likelihood of breaches: Assess the probability of a specific threat occurring based on historical data.
- Evaluate security risk levels: Identify the most vulnerable areas and prioritize protection measures.
- Support decision-making: Provide accurate information to security teams for informed actions.

Applications of Probability and Statistics in Cybersecurity

Probability and statistics are used across various fields, such as:

- Security risk analysis: Estimating potential damage from different types of threats.
- Intrusion Detection Systems (IDS): Detecting suspicious network activities using statistical data.
- User behavior analysis: Identifying unusual behaviors that may indicate breaches or threats.
- Spam classification: Using probabilistic models to detect harmful messages.
- Evaluating the effectiveness of security systems: Measuring the efficiency of security tools and making improvements.

Practical Applications of Probability and Statistics in Cybersecurity

Probability and statistics play a key role in practical cybersecurity applications, including:

- Detecting abnormal network patterns: Using statistical data to identify unusual or suspicious activities.
- Analyzing malware: Studying the behavior of malicious software to understand its operation and predict its impact.
- Predicting future attacks: Employing probabilistic models to estimate the likelihood of new threats.
- Analyzing login and logout logs: Monitoring data to detect unauthorized access attempts or suspicious activities.

Everyday Life Examples

The importance of probability in cybersecurity also appears in daily life, such as:

- Estimating the likelihood of personal account breaches: Analyzing risks associated with online accounts.
- Evaluating password security: Using probabilistic models to assess password strength.
- Understanding risks of using public Wi-Fi: Estimating the probability of being hacked on unsecured networks.
- Analyzing the likelihood of data leaks: Assessing the risk of sensitive data loss and taking preventive measures.

Linking Probability and Statistics with Computing

Probability and statistics are closely connected to computing and are widely used in various technical fields, including:

- Big data analysis: Processing massive amounts of data to extract valuable patterns and insights.
- Artificial intelligence (AI): Relying on probabilistic models to predict outcomes and make intelligent decisions.
- Machine learning: Training algorithms using statistical methods to determine likely outcomes and forecast future behaviors.
- Information security and cryptography: Using probability to assess security risks and design more effective protection systems.

Programming Applications in Probability and Statistics

Programming languages like Python and R are widely used for analyzing security data, helping to:

- Calculate the likelihood of attacks: Estimating risks based on historical data.
- Analyze failed login attempts: Detecting suspicious activities by studying login and logout logs.
- Detect abnormal behavior: Identifying unusual patterns in networks or user activities to prevent breaches.

Probability

Example Probability of Password Breach

$$\text{Formula: } P(E) = \frac{\text{NUMBER OF FAVORABLE OUTCOMES}}{\text{TOTAL NUMBER OF OUTCOMES}}$$

Example: A password consists of 4 digits (0–9).

What is the probability that an attacker guesses the password correctly on the first try?

Solution:

Total possible cases = $10 * 10 * 10 * 10 = 10,000$

Favorable cases = 1

$$P(\text{BREACH}) = \frac{1}{10,000} = 0.0001$$

Cybersecurity application:

Longer passwords → more possibilities

More possibilities → lower chance of breach

→ Therefore, long and complex passwords are used for account protection.

Statistics

Example: Analyzing Login Attempts

Formula (Arithmetic Mean): $\text{Mean} = \Sigma x / n$

Example:

Failed login attempts over 5 days:

Day 1: 2

Day 2: 3

Day 3: 2

Day 4: 3

Day 5: 20

Solution: $\text{Mean} = \frac{2 + 3 + 2 + 3 + 20}{5} = 6$

Σx = number of failed login attempts per day

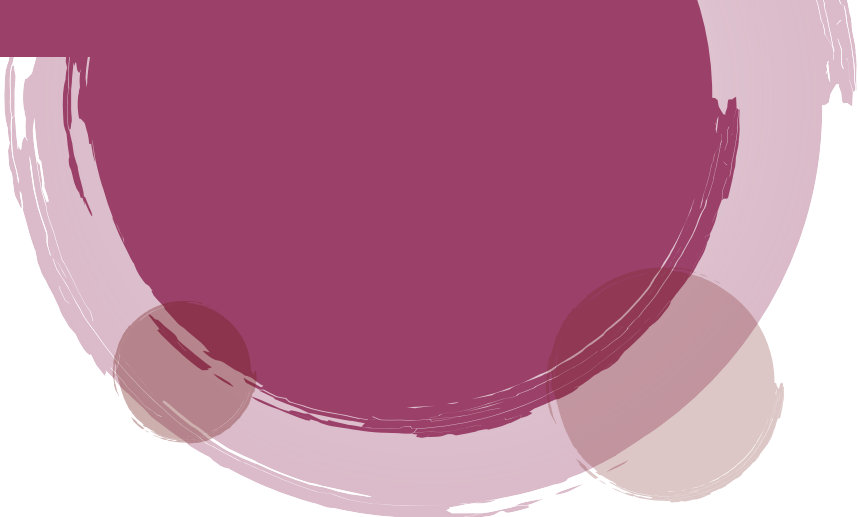
n = number of days

Normal mean $\approx 2-3$; Day 5 shows abnormal increase.

Cybersecurity application:

System suspects a breach attempt

Triggers alert to monitor suspicious activity



Conclusion

In conclusion, probability and statistics are considered fundamental pillars in the field of cybersecurity due to their significant role in understanding and analyzing data in a scientific and accurate manner. They contribute to detecting abnormal patterns, predicting potential cyberattacks, and assessing the level of risks that digital systems may face. Additionally, they support data-driven security decision-making, improve the efficiency of protection systems, and enhance incident response capabilities. Through the use of statistical methods, organizations can strengthen their ability to protect sensitive information, reduce losses caused by cyber threats, and achieve a higher level of security and stability in the digital environment

